



Critical heat flux prediction for annular channel through application of machine learning techniques

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ABSTRACT

Critical heat flux (CHF) is a prominent parameter for ensuring the security and economic efficiency of boiling heat transfer applications. Accordingly, the prediction of CHF is of vital importance in improving heat transfer performance and reducing the risk of failures. This study introduces a machine learning-based method for predicting flow boiling CHF in annular channels. A comprehensive dataset was established by combining the experimental data on the home-built experimental loop and the previous studies' data. Five different algorithms were employed to improve prediction accuracy: backpropagation neural networks, support vector machines, random forests, radial basis function neural networks, and convolutional neural networks. All predictive models utilized this comprehensive database, which included six parameters, for training and testing purposes. The results show that the machine learning models can effectively predict CHF, with support vector machines exhibiting the best performance, attaining a root mean squared error (RMSE) of 0.18558 and demonstrating excellent generalization ability. Then, feature importance analysis and sensitivity analysis were conducted to identify the most significant feature in predicting flow boiling CHF. This study presents a new approach to predict flow boiling CHF meanwhile offering some referable insights for the application of artificial intelligence in boiling heat transfer systems.

1. Introduction

Critical heat flux (CHF) is a critical parameter for ensuring the reliability and economic efficiency of the boiling heat transfer applications, e.g., nuclear power plants, and electronic cooling. CHF is defined as the heat flux at which a boiling crisis takes place, marking the transition from nucleate boiling to film boiling on the heating surface. Accurate prediction of critical heat flux is crucial for assessing the thermal-hydraulic behavior and enhancing operating efficiency and economy in the boiling heat transfer system.

Numerous factors can influence critical heat flux, including system pressure, mass flux, inlet subcooling, and geometric configuration, as demonstrated in Bruder et al. [1]. While traditional empirical correlations and mechanistic models provide foundational tools, their accuracy can be limited under wide operating ranges or for complex geometries like annular channels, necessitating the exploration of data-driven alternatives.

Over the past few years, the swift progression of computer technology and machine learning algorithms has opened up new avenues

for CHF prediction. For instance, algorithms such as support vector machines (SVM), multi-layer perceptions (MLP), gaussian process regression (GPR), convolutional neural networks (CNN), and artificial neural networks (ANN) have been increasingly implemented to CHF prediction. Machine learning (ML) techniques, as alternatives to look-up tables (LUTs) and empirical correlations, do not require a deep understanding of complex physical processes and mathematical theories. After training, ML models can effectively predict relevant parameters. Sallehuddin [2] achieved a prediction accuracy of 88.28% using DLNN and CNN algorithms within the ANN framework, combined with the Groeneveld LUT database. Grosfilley [3] compared three algorithms to predict the CHF using the LUT database. Of the algorithms studied, the neural network exhibited the greatest efficacy, yielding a standard deviation of 12.3% for the predicted versus measured CHF factor. This represents a three-fold improvement over the LUT method. He [4] explored a CHF prediction strategy based on sparsely distributed data points using the v-SVM algorithm, achieving high prediction accuracy in the pressure and mass flux parameter space. Lu [5] developed

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Nomenclature

Bo	Boiling number
C_p	Specific heat of liquid at constant pressure [J/kg/K]
D	Inner diameter of round tube [mm]
G	Mass velocity [kg/m ² s]
g	Acceleration due to gravity [m/s ²]
h_{fg}	Latent heat of evaporation [J/kg]
h_{sat}	Saturated enthalpy of vapor [J/kg]
Δh_i	Enthalpy of inlet subcooling [J/kg]
Δh_{sub}	Enthalpy of inlet subcooling [J/kg]
K	Parameter for inlet subcooling effect
L	Length of heated rod or tube [m]
L_B	Boiling length [m]
L_c	Axial distance between tube entrance and location of boiling crisis [m]
L_E	Effective length of tube [m]
n	Exponent of Y
p	Absolute pressure [MPa]
q_c	Critical heat flux [W/m ²]
q_{c0}	q_c for $\Delta h_i = 0$ [W/m ²]
q_{CHF}	Critical heat flux [W/m ²]
p_r	Relative pressure [MPa]
X	Vapor quality
X_c	X at location of CHF
X_{IE}	Effective inlet quality
X_{IN}	Inlet quality
Y	Shah's correlating parameter

Greek symbols

ρ_l	Density of liquid [kg/m ³]
ρ_v	Density of vapor [kg/m ³]
σ	Surface tension [N/m]
μ_L	Dynamic viscosity of liquid [Pa·s]
μ_G	Dynamic viscosity of gas [Pa·s]

neural network models based on neuromorphic-simulated event data to predict the onset of boiling crises. These predictive models can identify the impending boiling crisis before it causes damage to boiling surfaces, and event data can be successfully utilized in real-time, motion-dependent scenarios. Kim [6] investigated the effect of each parameter on CHF in a narrow rectangular channel and used the CHF correlation to generate the training data, and obtained an average neural network error of 3.65% after training, and the RMSE was 17.17%. Kumar [7] used various machine learning models to forecast the location of CHF, then chose the database of Becker [8] for model training and testing, and finally the ANN algorithm performed well. The ANN algorithm performed well with a MAPE of 6.04%. Yang [9] developed a new two-stage machine learning algorithm that can effectively identify the primary dimensionless numbers governing critical heat flux. This approach overcomes the shortcomings of traditional dimensional analysis and demonstrates superior generalization performance when tested on a much larger dataset. Jiang [10] predicted CHF based on the GPR-ACO and SVR algorithms, and observed that the GPR-ACO hybrid model outperforms the SVR, and both models are effective in improving the accuracy of CHF prediction. Rassoulinejad-Mousavi [11] employed image-based deep learning models to detect critical heat flux based on pool boiling images, and convolutional neural networks and transfer learning were trained to perform the prediction. The results demonstrate the ability of TL for handling data scarcity in pool boiling

with potentials for real-time implementation. Li [12] developed a CHF prediction model using a BP neural network. To assess the sensitivity of various parameters, he calculated the sensitivity coefficients by determining the partial derivatives of the prediction model. The results demonstrated that the BP neural network effectively predicted CHF, aligning closely with the experimental data. The prediction errors were within a 5% margin. Zhao [13] employed a hybrid framework that incorporates physical information with machine learning techniques to enhance predictive capabilities. A comprehensive study utilizing a vast CHF database specifically focused on DNB (Departure from Nucleate Boiling) demonstrated the hybrid framework's adaptability and resilience. The framework's ability to effortlessly expand its scope and complexity underscores its superior flexibility and robustness. Cabarcos [14] investigated the application of supervised learning algorithms to forecast temperatures in an experimental test setup. For CHF classification, the SVM model reached a remarkable F1-score of 0.872 on the testing dataset for CHF classification, demonstrating its superior performance. In terms of temperature prediction, the artificial neural network outperformed the other models, demonstrating a lower normalized mean absolute error (nMAE) of 6.18% on the testing dataset.

The majority of prior studies have been conducted based on Groeneweld's Look-Up Table (LUT) database [15], which offers CHF values calibrated for pipe configurations. While Groeneweld's LUT is primarily calibrated for pipe geometries, various studies have adapted or extended its principles and correlations to annular channels. However, these adaptations may still be limited by their specific calibration datasets or underlying assumptions, which our data-driven ML approach aims to overcome. While previous studies, such as those by Katto [16] and Bowring [17], have proposed numerous predictive models for CHF, these relationships remain constrained by narrow applicability to specific geometries, fluids, or flow conditions. Though mechanistically accurate, traditional models are computationally intensive and challenging to implement. In contrast, machine learning models provide a computationally efficient alternative for real-time predictions, capable of generalizing across broader parameter ranges. ML methods excel at identifying complex, non-linear relationships within large datasets, uncovering patterns that theoretical frameworks alone may overlook.

Therefore, there is a need for comparative evaluations of diverse ML models using comprehensive datasets for this specific geometry, coupled with analyses that elucidate parametric importance and quantify predictive uncertainty.

Addressing this gap, the present study aims to develop and evaluate robust machine learning models for predicting flow boiling CHF in annular channels. This is achieved by: (1) Establishing a comprehensive database for annular channel CHF by integrating new experimental data from a home-built facility with curated data from previous literature, covering an extensive range of operating condition parameters. (2) Systematically employing and comparing five distinct ML algorithms in combination of cross-validation and hyperparameter tuning techniques. (3) Conducting feature importance and sensitivity analyses to identify and understand the impact of key operating parameters on CHF in annular channels and performing the uncertainty quantification analysis to assess the confidence in the predictions. (4) Employing the comparative evaluation with empirical correlations to validate the model's accuracy and reliability.

This work provides a benchmark for ML-based CHF prediction in annular channels and offers insights into the application of AI for designing and operating safer and more efficient boiling heat transfer systems involving such geometries.

2. CHF database

2.1. Database construction

Before establishing the database, a brief discussion was conducted on the parameters relevant to CHF that need to be collected. According

Table 1
Ranges of experimental CHF dataset.

Author	D [mm]	L [mm]	P [MPa]	G [kg/m ² s]	ΔT_{sub} [K]	CHF [MW/m ²]
Kirby [18]	18.9–19.1	142–152	0.172	271.2–2278.5	3.81–46.01	1.325–4.418
Fiori [19]	4.6	179–254	0.148–0.510	542.5–2034.3	72.4–118.46	1.779–6.372
Molen [20]	43	190	0.100–0.200	1000.0–2500.0	14.21–59.41	2.500–4.000
Rogers [21]	9.0–17.0	480	0.156	55.0–650.0	42.53–92.53	0.250–1.450
Chun [22]	9.8	1842	0.570–5.850	550.0–650.0	16.94–20.31	0.900–1.220
Hass [23]	3.5–8.5	326	0.110–0.300	250.0–1000.0	39.50–73.29	0.964–2.400
Schoesse [24]	12.0	1000	0.128	15.5–258.7	7.00–52.00	0.191–0.720
Mayer [25]	10.3	454	0.110–0.226	51.5–106.5	3.00	0.478–0.637
Chen [26]	17.6	390	0.110–0.168	65.7–131.8	16.90–47.20	0.562–0.850
Hibiki [27]	5.6	550	0.500–1.000	17.7–130.7	51.80–119.90	0.215–0.929

to the paper [1], the incidence and severity of CHF are largely determined by the system's operating parameters. These parameters can be broadly categorized into three groups: thermal-hydraulic, geometric, and external. Among the thermal-hydraulic parameters, system pressure, mass flux, and local enthalpy/subcooling are of primary interest. Geometric parameters such as channel diameter, surface roughness, and heater thickness, and external parameters like heat flux distribution and dissolved gases are commonly considered crucial for subcooled flow boiling. Due to the assumption of uniformly heated surfaces and the lack of gas solubility in many mechanistic models for subcooled flow boiling, external influencing factors are omitted in this paper for conciseness.

Based on the above discussion, an all-inclusive literature survey was carried out to assemble experimental data on CHF in annular channels with water as the working fluid. As presented in Table 1, the dataset comprising 326 data points was compiled, encompassing six key parameters: hydraulic diameter(D), heated length(L), pressure(P), mass flux(G), inlet subcooling(ΔT_{sub}), and critical heat flux. Table 1 presents an epitome of the collected data, detailing the range of values for each feature. This dataset, derived from a collection of extensive studies, serves as the foundation for developing a predictive model for CHF. Notably, in all the literature reviewed for data collection, the annular channels were configured vertically for heating, consistent with our experimental arrangement. This consistency eliminates biases from flow regime variations caused by orientation-dependent phenomena such as buoyancy-driven stratification or asymmetric boiling. By restricting the dataset to vertical heating, we ensure model predictions remain physically coherent within the reactor-relevant design space. This intentionally narrow scope avoids overgeneralization risks while providing a robust foundation for future extensions to other geometries.

While our composite dataset combines in-house experiments and literature sources, its size remains constrained compared to the complexity of annular flow boiling systems. This limitation arises from the scarcity of publicly available annular channel CHF data, particularly for prototypical nuclear reactor conditions. The current dataset prioritizes conventional CHF-driving parameters (e.g., mass flux, pressure, subcooling). Other potentially influential factors, such as surface roughness or axial heat flux profiles, were excluded due to inconsistent reporting across studies.

2.2. Data analysis

Based on the collected dataset, a Pearson correlation analysis was carried out to explore the connections among various variables. The results are presented in Fig. 1. The Pearson correlation coefficient is a statistical metric utilized to quantify the strength of linear correlation between two variables. It is calculated using the following formula:

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}} \quad (1)$$

In Eq. (1), r represents the Pearson correlation coefficient, x_i and y_i denote the values at the i th observation of variables x and y , respectively, and $\bar{x}$ and $\bar{y}$ are their respective means. The correlation

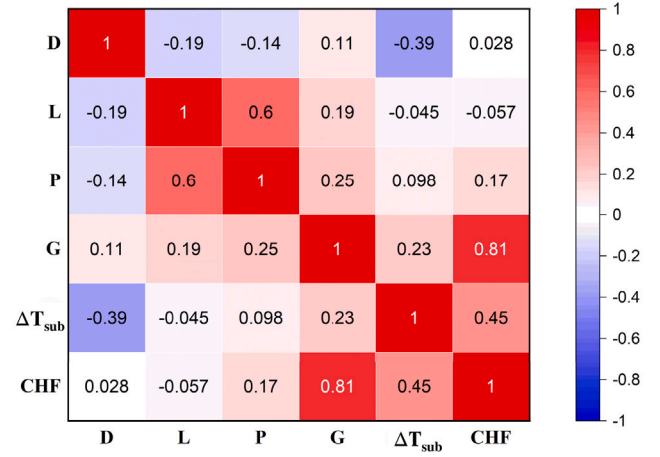


Fig. 1. Pearson correlation diagram.

coefficient r , a numerical value between -1 and 1 , quantifies the strength and direction of the linear relationship between two variables. A value closer to -1 or 1 indicates a stronger linear relationship, while a value closer to zero suggests a weaker or nonexistent one.

According to Fig. 1, a strong positive correlation was found between mass flux and CHF. Moreover, a moderate positive correlation was observed between inlet subcooling, system pressure, and CHF. In contrast, the correlation between hydraulic diameter and heating length, and CHF was negligible. Given the limited number of input features in the dataset, all five features were included in the training of the machine learning models. Fig. 2 presents the distribution of each parameter in the dataset, which demonstrates the completeness and consistency of the data.

2.3. Data preprocessing

Since the algorithms employed in this study are sensitive to the dimension of selected data, and the different features exhibit a spacious scope of values, we introduced data normalization prior to model training. By normalizing the data, we mapped the values of all features to the range of $[0, 1]$, ensuring that they share a common scale. By eliminating the influence of dimensions and orders of magnitude, this process enhances the model's predictive capabilities. The normalization process is defined by the following equation:

$$X' = \frac{X_i - X_{\min}}{X_{\max} - X_{\min}} \quad (2)$$

In Eq. (2), X_i represents the original data, X' denotes the normalized data, and $X_{\min}$ and $X_{\max}$ embody the minimum and maximum values of the original data, respectively.

In this research, we employed 10-fold cross-validation to assess the model's performance and mitigate the risk of overfitting [28]. This technique involves randomly splitting the dataset into 10 equal-sized

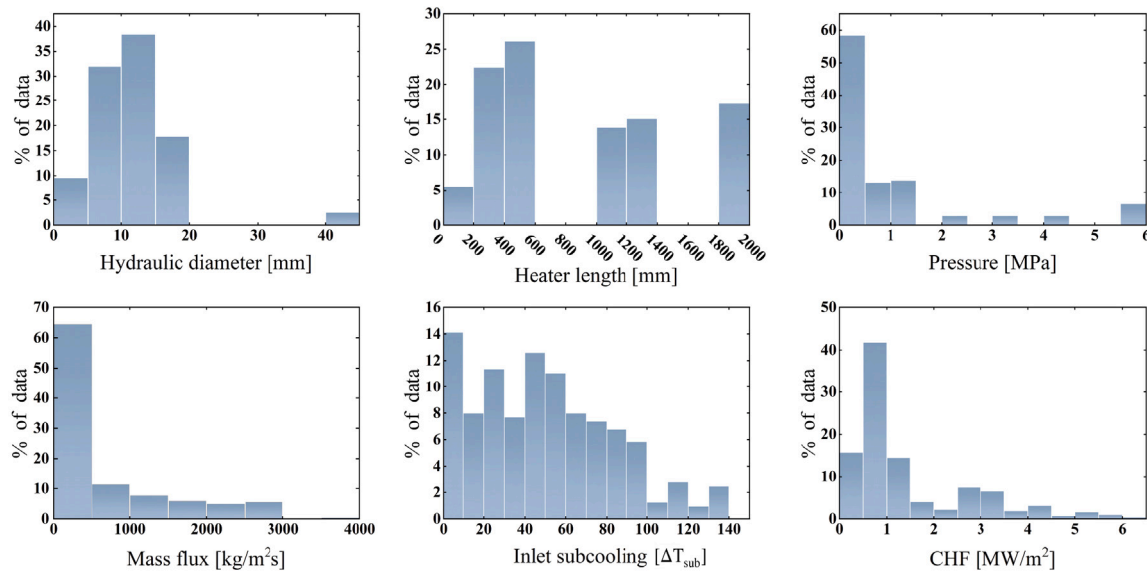


Fig. 2. Data distribution for each parameter.

portions. In each iteration, one portion is used for testing, while the remaining nine are used for training. By repeating this process 10 times and averaging the results, we obtain a reliable evaluation of the model's performance. This approach effectively reduces the randomness introduced by data splitting and improves the model's generalization ability.

3. Experimental research

3.1. Experimental loop

Based on the data acquired from the previous researches, in order to scale up the database, a custom-designed experimental loop was constructed to study the critical heat flux under various flow boiling conditions. As illustrated in Fig. 3(a), the loop is composed of a primary heating loop, a secondary cooling loop, a data acquisition and control system, and auxiliary systems. The primary loop, fabricated entirely from 304 stainless steel to prevent corrosion, is designed to operate at a maximum pressure of 10 MPa and a temperature of 275 °C. Deionized water is circulated as the working fluid. All the test points pertaining to pressure, flux and temperature are presented in Fig. 3(a). Fig. 3(b) shows the photograph of the experimental apparatus.

3.2. Test section

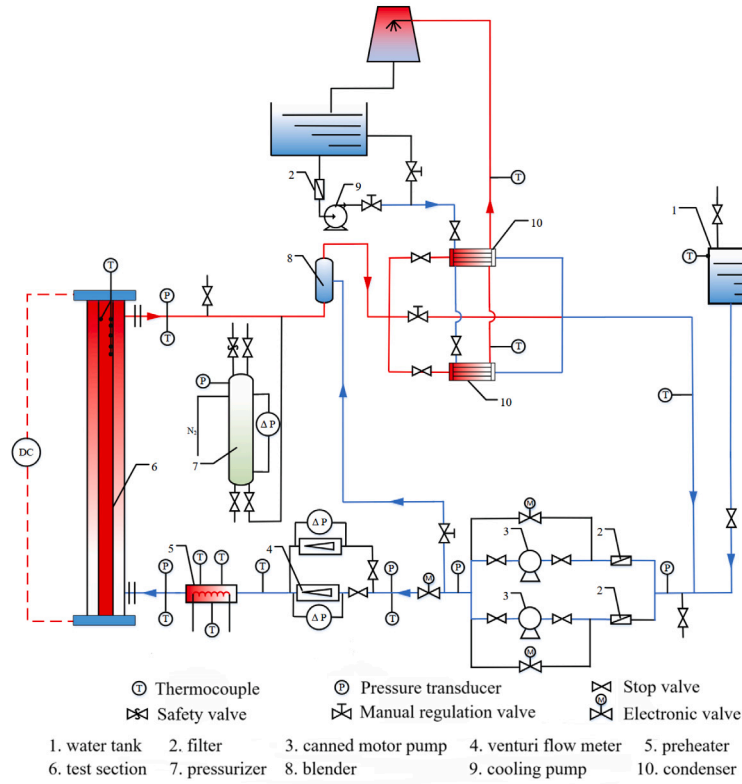
Conducting detailed thermal-hydraulic experiments on full-scale fuel bundle channels presents significant challenges due to limitations in experimental facilities, heating techniques, measurement technologies, and experimental costs. While larger bundle configurations can more accurately reflect the thermal-hydraulic performance of actual reactor cores, they are associated with complex experimental setups, high costs, and measurement difficulties. In contrast, lattice channel experiments offer several advantages. They simplify experimental procedures, reduce costs, and facilitate detailed measurements while still providing representative results. Moreover, visual experiments conducted in subchannels can provide direct insights into near-wall flow boiling behavior, aiding in the elucidation of boiling crisis triggering mechanisms and the development of predictive models. Therefore, this study selects a lattice channel with a geometry and thermal-hydraulic characteristics similar to those of a pressurized water reactor core component as the experimental test section.

The test section, shown in Fig. 4(a), is a vertical annular channel with a heated length of 1344 mm. The inner wall of the channel is heated by a cylindrical stainless steel heating rod with a diameter of 9.5 mm, while the outer wall is cooled by the flowing fluid. A transparent viewing window is installed to allow for visual observation of the flow boiling phenomena. To ensure accurate measurement of pressure and temperature, various sensors were strategically placed along the test section. Fig. 4(b) shows the photograph of the test section.

3.3. Experimental procedure and results

The experimental fluid is degasified before the start of the CHF test. To adequately degas the working fluid, the water tank heater is operated without flow for approximately five hours before each test, followed by circulating the coolant in the loop for several hours. Once the desired mass flow and subcooling levels are attained, the heating power supply is engaged, incrementally raising the heating power until a boiling crisis, characterized by a spike in wall temperature, is reached. Simultaneously, the data collection system is initiated, and adjustments are made to the mass flow in the secondary cooling system of the auxiliary system and the preheater power to maintain nearly constant inlet subcooling. The average heat flux is incrementally raised using slightly larger power increments, and as it nears the CHF, the power increment is decreased until a temperature spike is detected. Upon detection of a wall temperature spike, 30% of the power is promptly shut off. This indicates the completion of an experimental run. Following this, adjustments are made to the inlet subcooling, mass flow, and system pressure to achieve the desired subsequent conditions (refer to [31] for specifics). The experiments are carried out using various thermal hydraulic parameters. The ranges of experimental parameters for the inlet subcooling (ΔT_{sub}), mass flux (G), and system pressure (P) are outlined in Table 2, along with the experimental results.

The experimental measurement uncertainties have been fully documented in earlier publications [31]. Fluid enthalpy at the inlet and outlet is derived from temperature and pressure measurements, referencing the NIST properties database. Using the average of multiple single-phase heat balance experiments, heat loss is assessed and found to be less than 8% in this study. Repeated measurements yield uncertainties of 0.31% for the outer diameter and 0.37% for the length



(a) Experimental loop schematic



(b) Photo of experimental loop

Fig. 3. Experimental apparatus for the established loop [29,30].

of the heating rod. Uncertainties for mass flux (0.35%), fluid temperature (1%), wall temperature (1.67%), pressure (0.06%), DC voltage (0.1%) and DC current (0.5%) are derived from standard calibration instruments.

The above are all directly measured uncertainties, while this paper mainly focuses on the indirect measurement uncertainty of wall heat flux measurement. The wall heat flux can be determined based on the measured voltage, current, length, and outer diameter of the heating rod in the experimental setup. Therefore, according to the error propagation formula, the uncertainty of the wall heat flux is given

by:

$$\begin{aligned}
 \sigma_{q_w} &= \sqrt{\sum_{i=1}^n \left(\frac{\partial f}{\partial x_i} \right)^2 \sigma_{x_i}^2} \\
 &= \sqrt{\sigma_U^2 + \sigma_I^2 + \sigma_{Dr}^2 + \sigma_L^2} \\
 &= \sqrt{(0.1\%)^2 + (0.5\%)^2 + (0.31\%)^2 + (0.37\%)^2} \\
 &= 0.7\%
 \end{aligned} \tag{3}$$

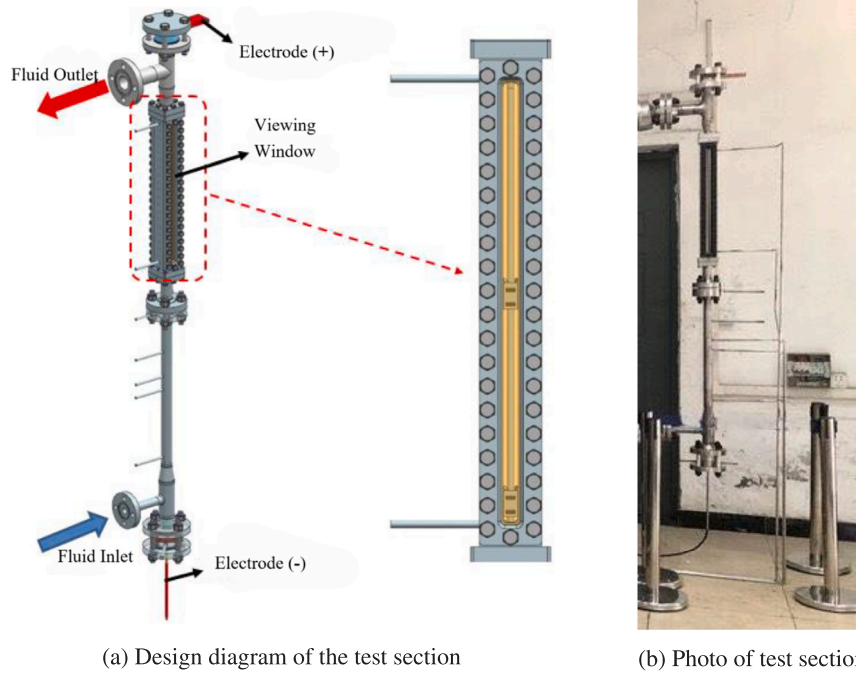


Fig. 4. Test section schematic [31].

Table 2
Current experimental data.

D [mm]	L [mm]	P [MPa]	G [kg/m ² s]	ΔT_{sub} [K]	CHF [MW/m ²]
10.9	1344	0.1 – 4	700 – 3560	40 – 90	1.11 – 4.19

4. Methodology: Machine learning models

With the intention of better assessing and evaluating predictive performance, five machine learning models were selected to make predictions on the aforementioned dataset using Python with scikit-learn and TensorFlow. These models exhibit strong nonlinear fitting capabilities and are well-suited for data regression prediction. This section provides a brief overview of these models and elucidates the hyperparameter optimization process during model construction.

4.1. Rationale for model selection

To ensure a comprehensive comparison, we selected models representing distinct categories of machine learning. The selection of these models was driven by a multi-faceted strategy aimed at exploring diverse learning paradigms, leveraging their individual strengths, and ultimately identifying the most suitable approach for this complex, multi-parameter prediction task. Each algorithm's technical details are provided in the following sections, this section focuses on the rationale for their inclusion in the context of CHF prediction.

4.1.1. Establishing a robust baseline (BPNN and SVM)

As a foundational and widely-used neural network architecture (see Section 4.2), BPNNs served as a crucial baseline model. Their ability to approximate any continuous function, given sufficient hidden units and appropriate training, makes them a versatile starting point for regression tasks. We anticipated that BPNNs would capture the overall non-linear relationships between the input parameters and CHF.

Support Vector Machines (detailed in Section 4.3) were included for their proven effectiveness in high-dimensional spaces and their robust generalization capabilities. SVMs, particularly with the use of kernel functions, are well-suited to handling non-linear relationships

and potentially noisy data, characteristics often present in experimental CHF datasets. The inherent focus of SVMs on maximizing the margin of separation (or tolerance in regression) was considered a valuable asset for achieving robust predictions.

4.1.2. Ensemble learning and feature importance (RF)

Random Forests (see Section 4.4) were selected for their ensemble learning approach, which combines multiple decision trees to improve predictive accuracy and reduce the risk of overfitting. The inherent randomness in data and feature selection during the training of individual trees contributes to the model's robustness. Furthermore, RF algorithms provide a built-in mechanism for estimating feature importance, which is crucial for gaining insights into the relative influence of different parameters on CHF - a key objective of this study.

4.1.3. Exploring localized and non-traditional approaches (RBFNN and CNN)

The inclusion of Radial Basis Function Neural Networks (RBFNNs, described in Section 4.5) was motivated by their distinct architecture, which utilizes radial basis functions as activation functions. Unlike the global nature of BPNN activations, RBFNNs exhibit a localized response, making them potentially more sensitive to specific combinations or ranges of input parameters. We hypothesized that this characteristic could be beneficial in capturing the intricate, localized variations in CHF that can occur due to subtle changes in parameters like mass flux and inlet subcooling. The often-faster training time of RBFNNs, compared to BPNNs, was also a practical consideration given the size of our dataset.

Convolutional Neural Networks (CNNs), typically associated with image and signal processing (see Section 4.6 for technical details), were incorporated as an exploratory, yet potentially insightful, approach. The main rationale for employing a 1D CNN was its potential to automatically discover and exploit local interactions or conjunctions between the input parameters that might be indicative of CHF conditions. Unlike fully connected networks where every input feature interacts with every neuron in the first hidden layer from the outset, a 1D CNN initially focuses on local groups of features (as defined by the filter size and the feature sequence). We hypothesized that certain

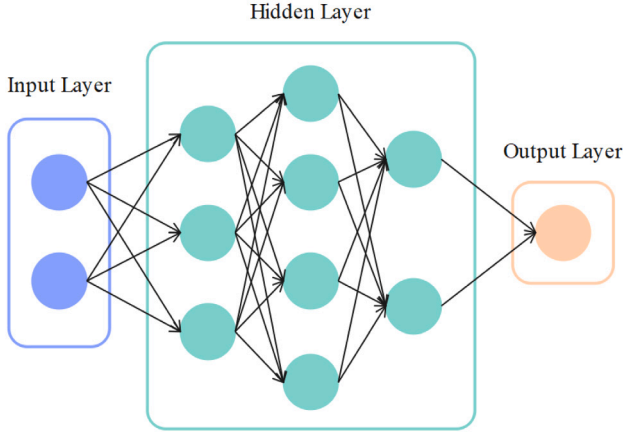


Fig. 5. Schematic of BPNN architecture.

critical heat flux phenomena might be triggered by specific combined states of a subset of parameters (e.g., a particular combination of mass flux ‘G’ and inlet subcooling ‘ ΔT_{sub} ’ when ‘Pressure’ P is within a certain range, if these features are ‘neighbors’ in the input sequence and covered by a filter). The CNN could learn to identify such predictive local interdependencies. Given the complex, non-linear nature of CHF, the inclusion of a 1D CNN was also partly exploratory. We aimed to investigate whether this architecture, commonly successful in other domains, could offer a novel perspective or capture predictive relationships in our tabular CHF dataset that might be less apparent to other conventional machine learning models. Its ability to learn problem-specific features was seen as a potential advantage.

4.2. Back propagation neural network (BPNN)

The BPNN is a type of multilayer feedforward neural network that is trained with the error back propagation method [32]. It is among the most commonly utilized neural networks. The back propagation neural network (BPNN) is grounded in the gradient descent method. This method aims to reduce the mean squared error between the network’s output and the desired output by calculating the gradient. The BPNN algorithm involves two primary steps: forward signal propagation and backward error propagation. During error calculation, the process moves from input to output, while weight and threshold adjustments are made from output to input. A back propagation neural network, as shown in Fig. 5, is structured with an input layer, one or more hidden layers, and an output layer. Each layer may have numerous nodes, and the connections between nodes across different layers are defined by weights. Non-linear transformations are applied to the nodes using activation functions.

4.3. Support vector machines (SVM)

SVM algorithms, parametric machine learning models, can address both linear and non-linear classification and regression problems [33]. These algorithms aim to find an optimal hyperplane in an N-dimensional space that clearly divides data points [34]. This hyperplane resides in an (N-1)-dimensional subspace of the N-dimensional feature space, as shown in Fig. 6. When tackling regression problems, SVMs construct a “margin” around the linear function $f(x)$ with a width of ϵ , often referred to as the tolerance. Samples falling within this margin incur no loss, meaning only support vectors influence the model. The superior model is obtained by minimizing the overall loss and maximizing the margin. Compared to traditional regression methods that consider a prediction to be correct only when $f(x)$ equals y exactly, SVMs allow for a certain degree of error, considering a

prediction correct as long as the deviation between $f(x)$ and y is within the specified tolerance. This flexibility makes SVMs more robust to noise and outliers. The objective of SVM on regression problems is to minimize $\frac{1}{2} \|\omega\|^2$. Additionally, points that lie within the boundary are required to fulfill the subsequent Eq. (4).

$$|y_i - (\omega - x_i + b)| \leq \epsilon \quad (4)$$

4.4. Random forest (RF)

The Random Forest model is an ensemble learning method that employs the Bootstrap Aggregating (Bagging) technique [35,36]. It can effectively solve nonlinear problems and is adept at handling large datasets and features. The Random Forest algorithm creates multiple decision trees and combines their predictions, either through averaging or weighted averaging, to arrive at the final prediction, as depicted in Fig. 7. By integrating multiple models, Random Forest effectively addresses overfitting and enhances the model’s generalization capability. The intrinsic mechanism of Random Forest lies in its randomness and ensemble nature. Randomness comes from random sampling of data and features, making each decision tree different and increasing the diversity of the model. The ensembling nature combines the predictions of multiple decision trees through voting or averaging, effectively reducing the variance of the model and improving its stability and accuracy.

4.5. Radial basis function neural network (RBFNN)

As a prominent category of artificial neural networks, RBFNN is characterized by a single hidden layer and based on function approximation [37]. The entire network is structured as feedforward networks with a three-layer architecture, as illustrated in Fig. 8. The input layer consists of source nodes that solely transmit data information without any transformation. The hidden layer, with a variable number of nodes depending on specific requirements, utilizes radial basis functions to transform input information spatially. The output layer reacts to input patterns, and its neurons employ linear activation functions to linearly combine the outputs of hidden layer neurons, producing the final neural network output. The most widely used radial basis function is the Gaussian kernel, expressed as:

$$R_i(x) = \exp\left[-\frac{\|x - c_i\|^2}{2\sigma_i^2}\right] \quad (5)$$

where c_i is the center of the i th basis function, having the same dimension as the input vector; $i = 1, 2, \dots, m$; x is an n-dimensional input vector; and σ represents the variance of the Gaussian function.

4.6. Convolutional neural network (CNN)

Convolutional Neural Networks are a type of feedforward neural network characterized by a convolutional architecture, which effectively minimizes the memory requirements associated with deep networks [38,39]. CNNs incorporate local connectivity and weight sharing, effectively reducing the number of network parameters and mitigating overfitting. Additionally, CNNs exhibit representation learning capabilities, enabling shift-invariant classification of input information based on their hierarchical structure. They are able to be applied to both supervised and unsupervised tasks.

To apply a CNN to our tabular dataset comprising N scalar input features, we conceptualized these features as a one-dimensional (1D) sequence (Fig. 9). Each data sample, originally a vector of N features $[f_1, f_2, \dots, f_N]$, was presented to the network as a 1D array of length N . This allows 1D convolutional filters to slide across this feature vector. While the intrinsic order of features in many tabular datasets is not as strictly defined as in time-series or image data, arranging them into a fixed 1D sequence allows the CNN to explore local relationships

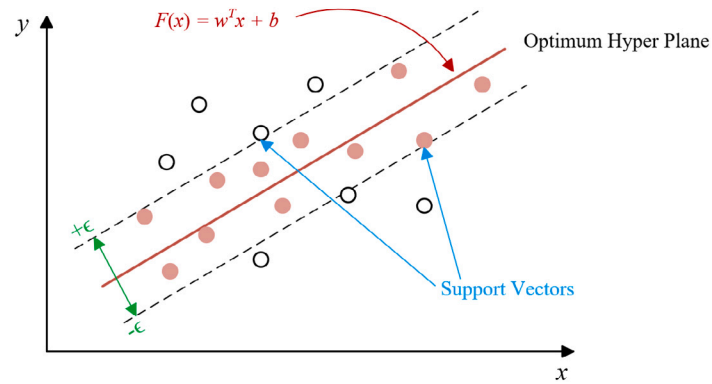


Fig. 6. Schematic of Support Vector Machine.

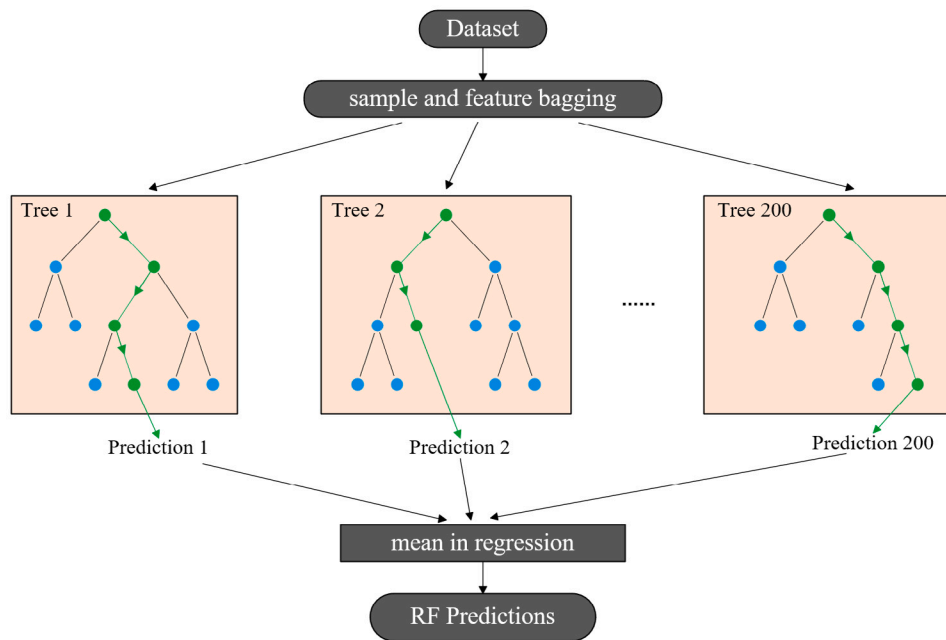


Fig. 7. Schematic of a Random Forest tree.

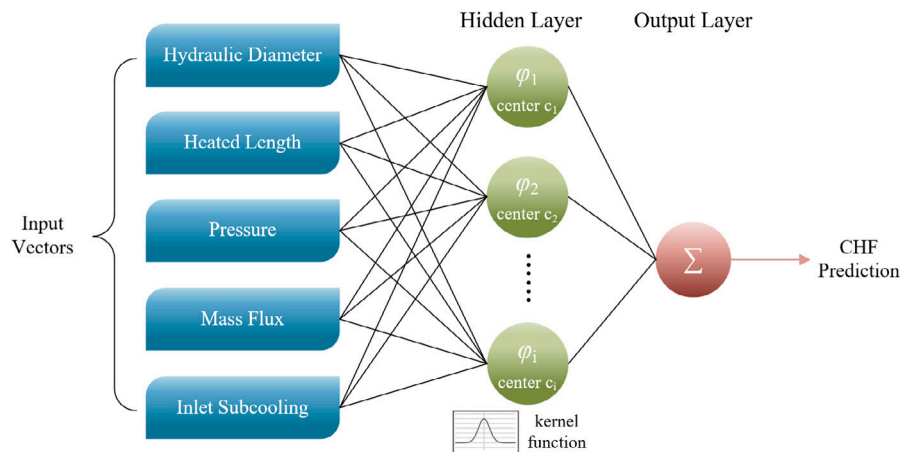


Fig. 8. Schematic of a RBFNN architecture.

between features as positioned in this sequence. For our study, the features [D, L, P, G, ΔT_{sub}] were consistently ordered for all input samples. The hypothesis is that meaningful interactions might exist

between ‘neighboring’ features in this defined order, or that the CNN can learn to identify important sub-groups of features irrespective of minor positional changes learned through its filters. One-dimensional

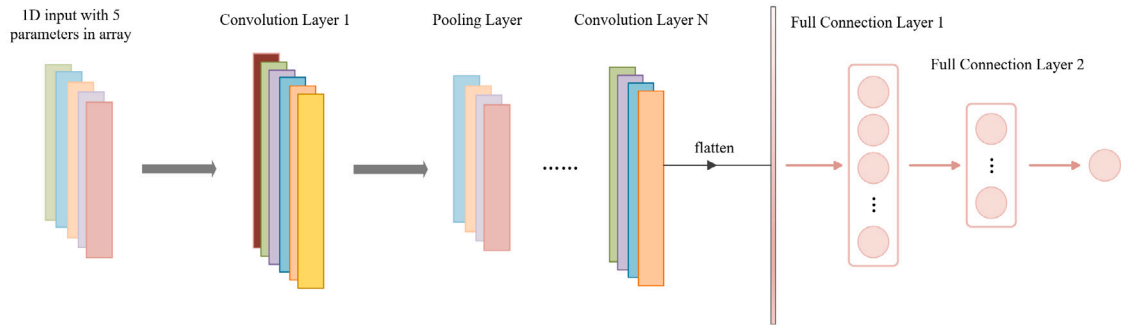


Fig. 9. 1D CNN regression architecture.

convolutional layers were then applied to this input sequence. A 1D filter (or kernel) of a predefined size (e.g., a kernel of size ‘k’ would cover ‘k’ adjacent features in the input sequence) slides across the feature vector. At each position, the filter performs a dot product, learning to detect specific local patterns or combinations of values among the ‘k’ features it currently spans. For example, a filter of size 3, when applied to features $[f_i, f_{i+1}, f_{i+2}]$, could learn to activate if a particular relationship exists between these three specific features. By using multiple filters in a convolutional layer, the network can learn to recognize various types of local feature interactions simultaneously. These learned ‘local patterns’ then become the inputs for subsequent layers, allowing the CNN to build a hierarchy of increasingly complex features derived from the initial input parameters.

4.7. Hyperparameters optimization

To achieve optimal performance, hyperparameters for all algorithms introduced in this section require tuning. In this study, Bayesian optimization [40] and grid search were employed for hyperparameter tuning. For algorithms with large hyperparameter search spaces, like BPNN, SVM, and CNN, Bayesian optimization was selected. Grid search was used for random forests and radial basis function neural networks. To improve the generalization ability of the models, 10-fold cross-validation was adopted for all optimization processes.

4.7.1. Bayesian optimization algorithm

Bayesian Optimization (BO) is a black-box optimization algorithm that iteratively constructs a probabilistic model to describe the posterior distribution of an objective function, gradually approaching the optimal solution. The core idea is to model the distribution of the objective function using Gaussian process regression and continuously update this distribution by adding sample points. The advantage of BO is that it can leverage information from previously searched points to find the global optimum with fewer iterations, thus improving both solution quality and search efficiency.

The objective function for the Bayesian Optimization was defined as minimizing the Root Mean Squared Error (RMSE) evaluated using k-fold cross-validation on the training dataset. A 10-fold cross-validation was employed to ensure robust evaluation of each hyperparameter combination and prevent overfitting during the tuning phase. The tuning results are presented in Table 3.

For the BPNN, 100 parameter iterations are deployed with the optimized hyperparameters including the number of hidden layers (N_h , searched within a range of 4 to 20), the learning rate (R, explored over 0.0001, 0.001, 0.01, 0.1), the training epochs (E, among 1000, 3000, 5000, 8000, 10000), and the training function (F, considering ‘trainlm’, ‘trainrp’).

For the SVM, through 70 parameter iterations, we tuned the kernel type including ‘linear’, ‘poly’, ‘rbf’, the regularization parameter C (searched in a range of 1 to 500), and the kernel coefficient gamma (for rbf kernel, in a range of 0.01 to 10)

For the CNN, the tuning process (300 iterations) encompassed the number of convolutional layers [1, 2, 3], the kernel sizes in each layer [4, 8, 16, 32, 64], the number of fully connected layers [1, 2], the number of neurons in each layers [16, 32, 48, 64, 96, 128], the mini-batch size (S, [32, 64, 128, 256]), the learning rate (R, over 0.0001, 0.001, 0.01, 0.1), the number of training epochs (E, among 400, 500, 600, 700, 800, 900, 1000), and the optimizer (Op, considering ‘Adam’, ‘RMSprop’, ‘SGD’).

The fundamental procedure of the BO algorithm are outlined as follows:

1. Define the objective function: This is the objective function to be optimized within the hyperparameter space, such as RMSE, or R^2 .
2. Define the search space: This is the space of hyperparameters to be optimized, like learning rate and the size of neurons in the hidden layer of a back propagation neural network.
3. Define initial hyperparameters: Set initial values for the hyperparameters to be optimized.
4. Generate initial hyperparameter combinations and evaluate: Generate a hyperparameter combination based on the initial values and calculate the corresponding objective function value.
5. Update the hyperparameter distribution: Update the distribution of hyperparameters based on the evaluated objective function value and generate a new hyperparameter combination.
6. Iterate: Repeat steps 4 and 5 until convergence or reaching the maximum number of iterations.
7. Return the optimal hyperparameters: Return the hyperparameter combination with the best objective function value.

4.7.2. Grid search algorithm

Grid search is a simple yet effective hyperparameter optimization method. It exhaustively searches through a predefined grid of hyperparameter values, evaluating the performance of each combination and selecting the one that yields the best performance metric, such as RMSE, MAE, or R^2 . While straightforward and easily parallelizable, grid search becomes computationally expensive and inefficient when the hyperparameter space is large and the number of hyperparameters is high. The outcome of the optimization process are illustrated in Table 4.

For the RF model, key hyperparameters optimized were the number of trees in the forest (Tree, [50, 100, 200, 300, 400, 500]), and the minimum number of samples required at each leaf node (Leaf, [1, 2, 3, 4, 5]).

The RBFNN optimization targeted two key parameters: the number of radial basis neurons ($5 \leq N \leq 30$) and the function spread parameter (range: 10–50).

Table 3
Hyperparameter tuning results using Bayesian Optimization algorithm.

Models	Hyperparameters
BPNN	$N_h = 8$, $R = 0.001$, $F = \text{trainbr}$, $E = 10,000$
SVM	kernel = rbf, $C = 185.1$, $\gamma = 1.6$
CNN	convolutional layer kernel 1 size = 8, convolutional layer kernel 2 size = 32, fully connected layer 1 size = 48, fully connected layer 2 size = 32, dropout = 0.16, $S = 128$, $R = 0.001$, $E = 700$, Op = 'RMSprop'

Table 4
Hyperparameter tuning results using grid search.

Models	Hyperparameters
RF	Tree = 200, Leaf = 1
RBFNN	$N = 15$, Spread = 38.7

Table 5
Performance comparison of implemented models.

Models	RMSE	MAE	R^2
BPNN	0.2117	0.1181	0.9673
SVM	0.1856	0.1029	0.9753
RF	0.2819	0.1541	0.9414
RBFNN	0.2219	0.1330	0.9643
CNN	0.2162	0.1275	0.9670

4.8. Model performance evaluation methodology

Machine learning algorithms construct probabilistic models to describe datasets, subsequently employing these models to generate predictions for previously unobserved data. To evaluate the prediction accuracy of these models, various metrics are employed. For regression tasks, commonly used metrics include root mean squared error (RMSE), mean absolute error (MAE), and coefficient of determination (R^2). The primary output parameter for evaluation is the critical heat flux.

RMSE is sensitive to outliers and is commonly used when the magnitude of errors is of primary concern, such as in financial forecasting. MAE, on the other hand, is less sensitive to outliers and is often preferred when the goal is to minimize the average magnitude of errors. R-squared quantifies the extent to which the variance in the dependent variable can be explained by the independent variables included in the model. Although R^2 is a useful indicator of model fit, it does not directly reflect the magnitude of prediction errors.

The mathematical expressions for the aforementioned metrics are as follows:

$$RMSE = \sqrt{\frac{1}{m} \sum_{i=1}^m (CHF_{prd,i} - CHF_{exp,i})^2} \quad (6)$$

$$MAE = \frac{1}{m} \sum_{i=1}^m |CHF_{prd,i} - CHF_{exp,i}| \quad (7)$$

$$R^2 = 1 - \frac{\sum_{i=1}^m (CHF_{prd,i} - CHF_{exp,i})^2}{\sum_{i=1}^m (CHF_{exp,i} - \bar{CHF}_{exp})^2} \quad (8)$$

where: m is the dataset size, $CHF_{prd,i}$ denotes the predicted value of CHF for the i th data point, $CHF_{exp,i}$ represents the experimental CHF value for the i th data point, and $\bar{CHF}_{exp}$ is the mean of the experimental CHF values.

5. Results and discussion

5.1. Model predictions and performance

All data was preprocessed following the method described in Section 2.4. Subsequently, hyperparameters for each algorithm were tuned, and model training and prediction were conducted using 10-fold cross-validation. All algorithms employed the same shuffled training and testing sets for regression prediction, facilitating performance comparison. As illustrated in Table 5, the prediction results of the four models were quite close, exhibiting excellent generalization ability. The RMSE values were all less than 0.3, and the R^2 values were all greater than 0.94, indicating high prediction accuracy. Among the models, SVM exhibited superior performance, achieving a mean RMSE of 0.18558 and an R^2 of 0.97525 after cross validation. This superior performance can be attributed to the fact that the optimization objective of SVM is to minimize structural risk rather than empirical risk. By introducing the concept of margin, SVM obtains a structural description of the data

distribution, reducing the dependence on data scale and distribution, and thus avoiding overfitting. Compared to other algorithms, SVM is more suitable for small sample datasets. Therefore, this model effectively improved the learning ability and generalization capability of the model, demonstrating notable predictive performance on the dataset.

As depicted in Fig. 10, the predicted values align closely with the experimental values, indicating a good fit. The scatter plot in Fig. 11 compares the predicted and experimental values for CHF. While some data points exhibit noticeable underestimation and overestimation, the overall trend remains consistent, and the data points closely align with the fitting line, suggesting high prediction accuracy. Fig. 12, the error histogram, shows that the majority of data deviations are within 20%, with only a small number of large positive or negative deviations. As shown in Fig. 13, the R^2 values for all iterations, save for the fourth and eighth, are remarkably close to 1. This strongly suggests that the model can effectively account for the observed data variability, demonstrating exceptional model fit.

Given the superior predictive outcomes of the SVM model, it was chosen as the optimal model for subsequent sensitivity analysis. The prediction accuracy of the model was evaluated using different dataset partitioning methods to determine its generalization ability. As shown in Table 6 and Fig. 14, the 10-fold cross-validation achieved the best performance, showing a significant improvement compared to the traditional single dataset partitioning method. This suggests that the model possesses robust generalization capabilities, mitigating the risk of overfitting. This is because cross-validation can obtain as much effective information as possible from limited data, avoiding the limitations and particularity of traditional fixed dataset partitioning, thereby obtaining a more reasonable and accurate evaluation of the model.

Having established the overall predictive accuracy, particularly the superior performance of the SVM model, the subsequent analyses aimed to understand the factors driving these predictions.

5.2. Discussion on parametric effects

5.2.1. Feature importance analysis

Although the Pearson heatmap indicates a linear correlation between the input features and the output variable, further investigation is necessary to evaluate the relative importance of the selected features in CHF prediction. Recognizing the prevalent nonlinearity in regression tasks, three feature importance models have been implemented using the current CHF database.

Among the five machine learning models employed to analyze the CHF datasets, only Random Forest provides the capability to assess feature importance. One of the metrics used for feature importance evaluation in Random Forest is the Gini index. The Gini impurity is a metric used to measure the impurity or heterogeneity of a dataset, and it can also be employed to evaluate the importance of features. In

Table 6
SVM sensitivity analysis.

Partitioning method	RMSE	MAE	R ²	Test Samples within $\pm 20\%$ error
10-fold CV	0.1856	0.1029	0.9753	94.48%
5-fold CV	0.2028	0.1052	0.9736	94.79%
80%train + 20%test	0.2756	0.1403	0.9562	85.94%
90%train + 10%test	0.3333	0.1809	0.9483	84.28%

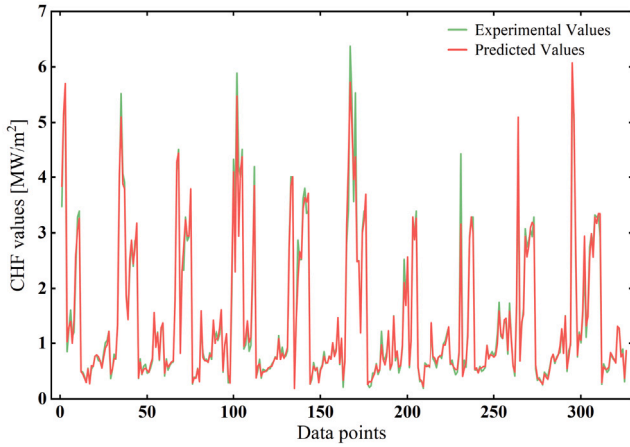


Fig. 10. Comparison of predicted and experimental CHF using SVM.

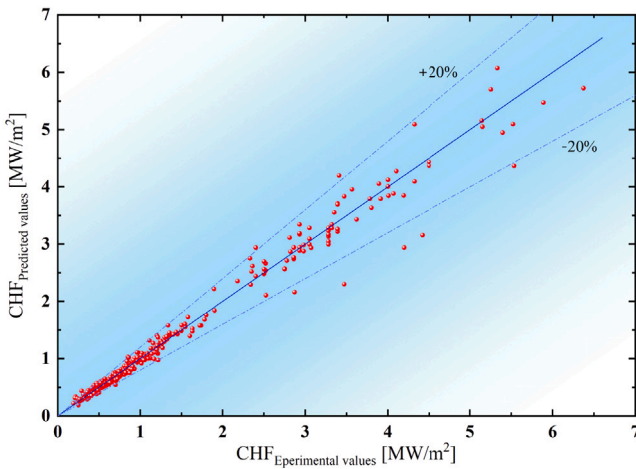


Fig. 11. Scatter plot of SVM result with best fit line.

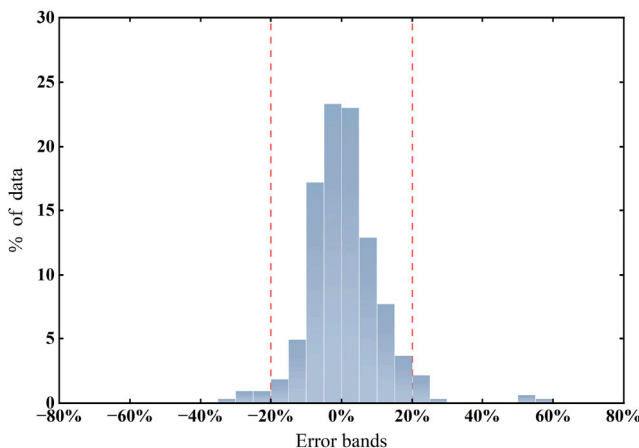


Fig. 12. Error histogram of SVM prediction.

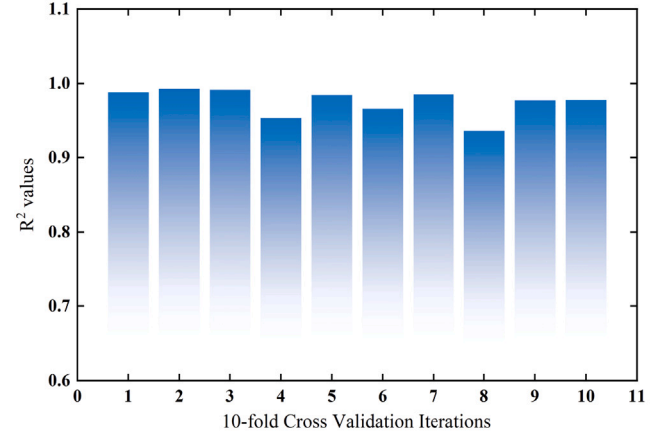


Fig. 13. The R² values in each iteration using SVM.

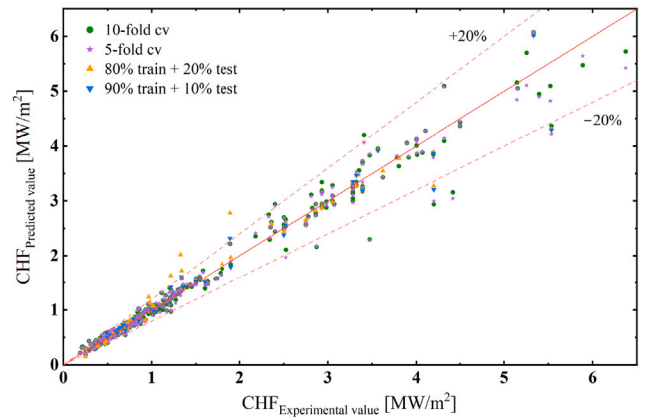


Fig. 14. Prediction results with different partitioning methods.

random forests, the Gini index is utilized to select the optimal feature for splitting nodes in a decision tree. By comparing the changes in Gini impurity when splitting on different features, we can assess the contribution of each feature to improving the purity of the model. A lower Gini impurity indicates that the feature plays a more substantial role in enhancing model purity, thereby indicating greater significance.

Permutation feature importance is a model inspection technique used to assess the individual contribution of each feature to the statistical performance of a trained model when applied to a given dataset. This approach is especially valuable for nonlinear or black-box models. By randomly permuting the values of a single feature and noting the subsequent decline in model performance, the model's reliance on that particular feature can be ascertained. A major advantage of permutation feature importance is its model-agnostic nature, meaning it can be applied to any fitted estimator. Furthermore, it can be computed multiple times using different permutations of features, providing an additional measure of the variance of the estimated feature importance for a specific trained model.

Recursive Feature Elimination (RFE) is an efficient feature selection method that recursively eliminates features until an optimal subset

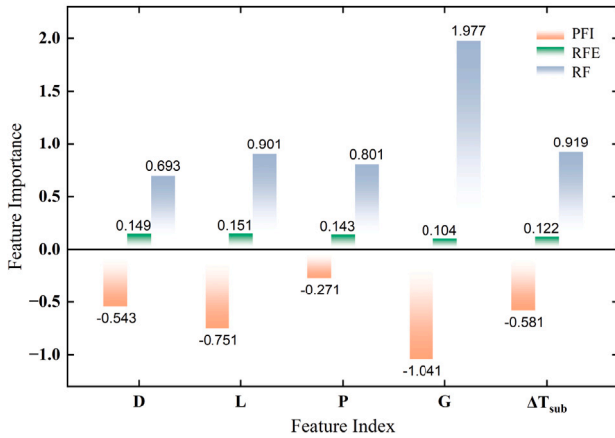


Fig. 15. Importance evaluation results.

is found. RFE operates by training a model with all features, then eliminating the least important features based on the model's inherent feature importance metric. The process is repeated until a specified number of features is reached or a stopping criterion is met. In RFE, a "lower ranking" indicates a more important feature. While the feature with the highest rank is deemed the least important, the feature ranked first is considered the most crucial. This ranking mechanism differs from direct feature importance scores (such as feature importance in random forests), where a higher score indicates greater importance. RFE iteratively removes features with the lowest weights or importance, retraining the model on the remaining features to obtain the ranking of each feature.

Feature importance was evaluated using the above methods (Fig. 15).

The paramount importance of mass flux (G) was unequivocally confirmed by three distinct feature evaluation techniques (PFI, RFE, and RF), highlighting its dominant role in CHF phenomena within annular channels. Beyond mass flux, both inlet subcooling (ΔT_{sub}) and heated length (L) consistently emerge as significant secondary parameters across multiple evaluation methods. While PFI suggests L has a slightly greater impact than ΔT_{sub} , RFE and RF indicate the reverse, underscoring the complex interplay and the challenge in definitively ranking these secondary effects without considering the evaluation methodology. The relative importances of pressure (P) and diameter (D) show more variability across the methods. While PFI indicates pressure as the least influential among the tested parameters, RFE suggests it has moderate importance, greater than diameter. Conversely, Random Forest highlights diameter (D) with a notable importance (0.693 in our analysis for RF), potentially ranking it above pressure for this specific model. This divergence suggests that the influence of P and D might be more model-dependent or captured differently by various algorithms, possibly due to non-linear relationships or interactions that each method weighs distinctly.

The analyses not only rank the features but also provide a quantitative sense of their relative impact. For instance, RFE suggests that mass flux (G) is over twice as influential as the next most important parameter (ΔT_{sub}), while PFI shows a more graded, though still dominant, influence of G. This quantification is a step beyond qualitative understanding.

By employing three distinct methods for feature importance evaluation, we offer a more robust interpretative framework. The consensus on the primary importance of G, ΔT_{sub} , and L provides high

confidence. The observed variations for D and P suggest these parameters may exhibit: (1) subtle nonlinear or interactive effects captured differently across methods, (2) importance shaped by dataset-specific characteristics, or (3) context-dependent behavior warranting targeted mechanistic investigation to explain method-dependent importance variations.

This framework will enable identification of specific operating regimes where current prediction methods may require refinement, while guiding the design of future targeted experiments to investigate the most sensitive parameter ranges and key interactions. Furthermore, it establishes a benchmark for developing computationally efficient, physics-informed surrogate models with greater sophistication.

5.2.2. Sensitivity analysis

Sensitivity analysis (SA) plays a pivotal role in evaluating the robustness and interpretability of machine learning models for CHF prediction. This section employs three complementary global sensitivity analysis methods – the Fourier Amplitude Sensitivity Test (FAST), One-at-a-time (OAT) and Variance-Based Sensitivity Analysis (VBSA) – to systematically quantify the relative importance of input features and their interaction effects on model outputs. The dual-method approach enhances methodological rigor by cross-validating sensitivity metrics through distinct mathematical frameworks.

The FAST method, originally developed by Cukier, utilizes Fourier series decomposition to disentangle input–output relationships. By assigning unique integer frequencies to each input parameter and applying spectral analysis to model outputs, FAST efficiently computes first-order sensitivity indices (S_i) that represent the fractional contribution of individual features to output variance. This approach demonstrates particular strength in handling nonlinear system responses while maintaining computational efficiency through its single-model-run implementation strategy.

The VBSA framework, grounded in Sobol's decomposition theorem, employs a Monte Carlo sampling scheme to decompose the total output variance into components attributable to individual parameters and their interactions. Through computation of first-order (S_i) and total-effect (S_{Ti}) indices, VBSA provides comprehensive insights into both main effects and higher-order interactions.

The sensitivity analyses were conducted using the Sensitivity Analysis For Everyone (SAFE) toolbox, an open-source framework specifically designed for robust implementation of global sensitivity methods. The toolbox's standardized workflow ensured methodological consistency between FAST and VBSA through three critical phases, including parameter space sampling, model interfacing and index computation. The SA results, as shown in Fig. 16.

Sensitivity analysis, employing FAST and VBSA techniques, revealed key insights into the factors influencing CHF. The results, shown in Fig. 16, underscore the dominant role of mass flux, with high sensitivity indices across all methods, aligning with the understanding that increased mass flux enhances heat removal from the heated surface. The moderate sensitivity of pressure and subcooling temperature reflects their contributions to stabilizing the two-phase flow and increasing the heat absorption capacity of the coolant, respectively. Furthermore, the hydraulic diameter, demonstrates influence on the interactions with the other features, but its direct influence on CHF is negligible. Conversely, the heated length shows little impact directly or interactively. These findings highlight the importance of controlling mass flux and operating conditions.

To further understand the influence of individual input features on the predicted Critical Heat Flux, a One-At-A-Time (OAT) sensitivity analysis was performed. The OAT method is a common technique used to assess the impact of input variables on a model's output. It involves systematically varying one input feature within its range while holding all other features constant at a baseline value. The resulting change in the model's output is then observed and plotted. This allows for a direct visualization of the relationship between each input feature and the

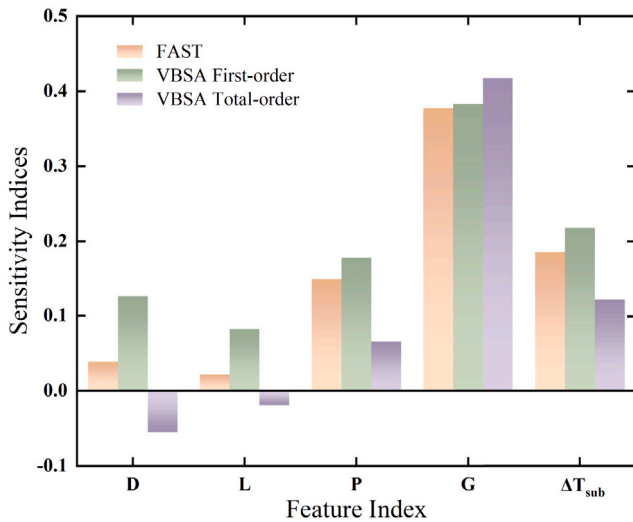


Fig. 16. Sensitivity analysis results.

predicted CHF, revealing the direction and magnitude of the feature's influence. While OAT is simple and intuitive, it is important to note that it does not capture potential interactions between features; it only assesses the individual effect of each feature in isolation.

Fig. 17 presents the results of the OAT sensitivity analysis for the six input features: D, G, L, P, and ΔT_{sub} . The observed trends provide insights into the potential underlying mechanisms governing CHF, as discussed below:

Feature D (Diameter): The model output shows a strong negative correlation with diameter. As D increases, the predicted CHF decreases significantly. This aligns with established CHF correlations and experimental observations. A larger diameter generally leads to a lower CHF because it increases the heated surface area, making it more difficult to remove the heat generated, and potentially leading to earlier vapor blanketing of the surface.

Feature L (Heated Length): The model output decreases as the heated length increases. This is consistent with the understanding that longer heated lengths provide more opportunity for vapor accumulation along the flow path. This accumulated vapor can coalesce and form a vapor blanket, leading to a premature CHF condition.

Feature P (Pressure): The predicted CHF initially increases with pressure, reaches a peak, and then decreases. This non-monotonic behavior is characteristic of the pressure effect on CHF. At lower pressures, increasing pressure enhances bubble departure frequency and reduces bubble size, improving heat transfer and increasing CHF. However, at higher pressures, the vapor density increases significantly, which can hinder bubble departure and lead to a reduction in CHF. The peak in the curve suggests the model has captured this complex interplay.

Feature G (Mass Flux): The predicted CHF increases significantly with increasing mass flux, exhibiting a near-linear positive relationship, although it starts to plateau at higher G values. This is a well-known and crucial factor in CHF. Higher mass flux enhances convective heat transfer, sweeping away vapor bubbles more effectively and delaying the onset of CHF. The plateauing suggests that at very high mass fluxes, other factors (e.g., pressure drop limitations) may begin to limit the further increase in CHF.

Feature ΔT_{sub} (Inlet Subcooling): The consistently increasing CHF with higher ΔT_{sub} reflects the enhanced heat sink capacity of the subcooled liquid. Higher subcooling means the liquid is further from its saturation point, allowing it to absorb more energy before significant vapor formation occurs. This delays the onset of the critical vapor blanketing that defines CHF. This overall trend confirms that

subcooling, by delaying and suppressing vapor formation, is a crucial factor in preventing the insulating vapor film that triggers the dramatic reduction in heat transfer characteristic of CHF.

Identifying the key parameters provides insight into the model's learning; however, understanding the confidence in its predictions is equally crucial.

5.3. Uncertainty quantification analysis

To address the critical need for uncertainty quantification (UQ) in CHF prediction, we implemented a method combining ensemble learning with quantile analysis, leveraging our best-performing SVM model.

An ensemble of SVM models was created using Bagging techniques which involved the following steps. From the original training dataset containing M data points, N bootstrap samples were generated. Each bootstrap sample was created by drawing M data points with replacement from the original training dataset. And then N individual SVM models were trained, one on each of the N bootstrap samples. Each SVM model utilized the optimized hyperparameter configuration (as determined in Section 4.7.1) to ensure that each ensemble member was itself a strong learner. For this study, an ensemble of N = 200 SVM models was created. This ensemble approach aims to capture model uncertainty arising from variations in the training data.

Once the ensemble of N SVM models was trained, it was used to generate predictions and quantify uncertainty for new, unseen data. For any given input data point from the test set, each of the N SVM models in the Bagging ensemble produced an individual CHF prediction. This resulted in a distribution of N predicted CHF values for that single input instance.

The final point prediction for CHF from the ensemble was taken as the median of these N predictions. The median is preferred over the mean as it is generally more robust to outliers or skewed prediction distributions from the ensemble members. To quantify the uncertainty associated with this median prediction, a 90% prediction interval (PI) was constructed from the ensemble's prediction distribution. This was achieved using the quantile method. Specifically, the lower bound of the 90% PI was defined as the 5th percentile of the N predictions, and the upper bound was defined as the 95th percentile of the N predictions. This interval $[P_{5th}, P_{95th}]$ is expected to contain the true CHF value with 90% confidence, based on the information captured by the ensemble.

The performance of the uncertainty quantification methodology is detailed in Fig. 18. This histogram illustrates the distribution of the calculated 90% prediction interval widths across the test dataset. The majority of PI widths (over 75% of samples) fall within approximately 0 to 0.5 MW/m². This indicates that for a substantial portion of the predictions, the ensemble model provides relatively tight uncertainty bounds, signifying a good level of confidence in these specific CHF estimates. The tail extending to the right (widths up to about 3.0 MW/m²) shows that for some instances, the model expresses higher uncertainty, which is an important characteristic of a well-calibrated UQ system.

This scatter plot (Fig. 18(b)) investigates the relationship between the width of the 90% prediction interval and the magnitude of the prediction error. There is a discernible positive correlation, although with considerable scatter. Points with very small residuals (highly accurate predictions) predominantly have narrow PI widths. As the absolute residuals increase, there is a tendency for the PI widths to also become larger and more dispersed. This trend is desirable as it suggests the UQ methodology is somewhat "aware" of the model's accuracy for a given prediction. When the model makes a larger error, the associated PI tends to be wider, appropriately reflecting increased uncertainty. This behavior is crucial for practical applications, as wider intervals can signal predictions that should be treated with more caution.

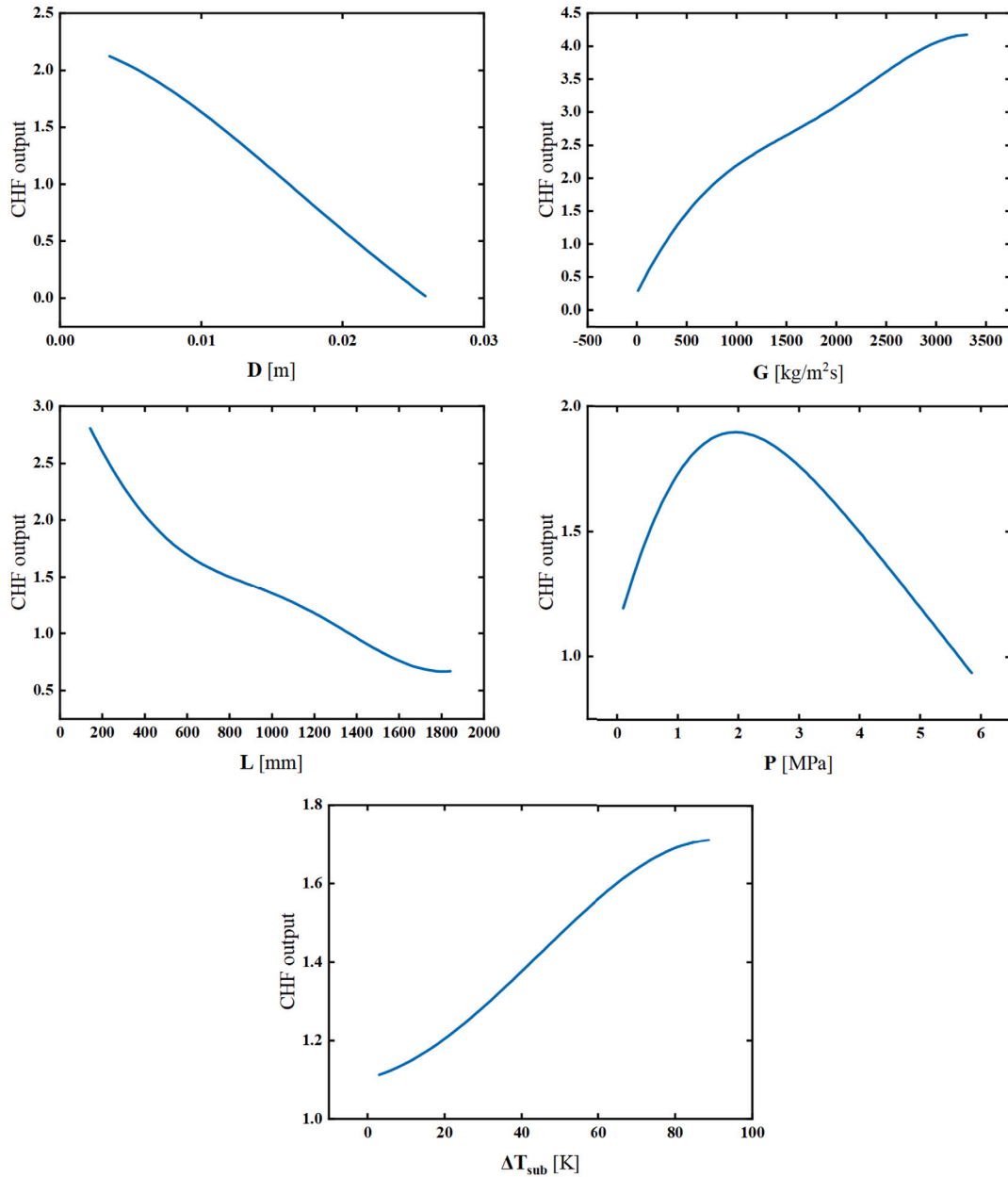


Fig. 17. OAT analysis results.

Fig. 18(c) examines how the uncertainty (PI width) varies with the magnitude of the predicted CHF value. A clear trend is visible where lower median predicted CHF values (below approximately 2.5–3.0 MW/m²) are generally associated with narrower PI widths. As the median predicted CHF value increases beyond this range, the PI widths tend to increase and exhibit greater variability, with some widths reaching up to 3.0 MW/m² for higher CHF predictions. This indicates that the model expresses more confidence (narrower PIs) when predicting lower CHF values and appropriately reflects greater uncertainty (wider PIs) for higher CHF values.

Fig. 18(d) shows the median predicted CHF values, and the 90% prediction intervals for data in the test set, sorted by the median predicted CHF. Crucially, this visualization supports the excellent quantitative result that the Prediction Interval Coverage Probability (PICP) is 90.6%. This means that 90.6% of the true experimental CHF values fall within the 90% prediction intervals generated by the model, which is very close to the nominal 90% target and indicates a well-calibrated

uncertainty assessment. Meanwhile, The plot visually confirms the trends seen in Figure Fig. 18(c), with narrower PIs typically observed at lower predicted CHF values and wider PIs at higher predicted CHF values. This plot provides strong evidence for the reliability of the UQ method. The high PICP value, coupled with the visual confirmation of coverage, demonstrates that the ensemble SVM model, along with the quantile analysis, can effectively generate meaningful and statistically sound prediction intervals.

With a comprehensive understanding of the ML model's performance and uncertainty, its practical utility was further evaluated by comparing it against established benchmarks.

5.4. Comparison with empirical correlations

To rigorously evaluate the predictive performance of the developed machine learning model, a comparison was conducted against two widely-used and highly-regarded empirical correlations for Critical

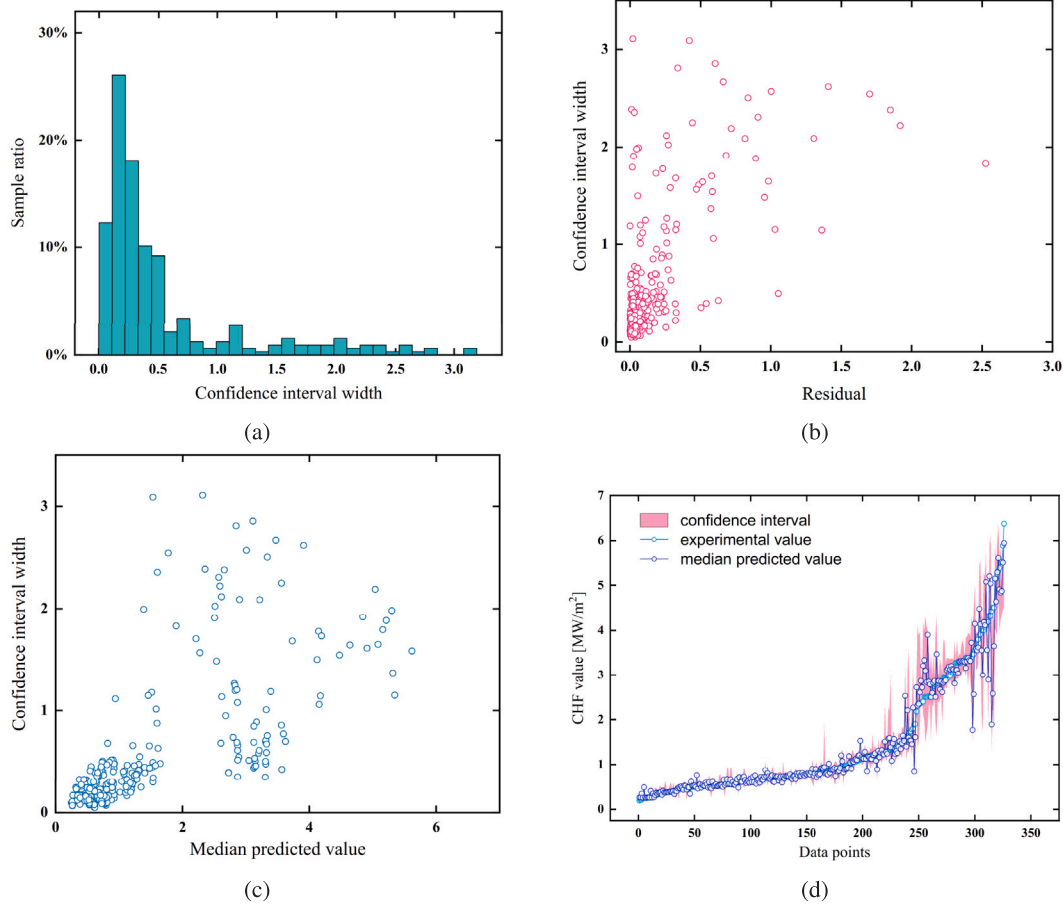


Fig. 18. Uncertainty quantification results for the ensemble SVM model using quantile analysis, (a) distribution of confidence interval widths, (b) confidence interval width vs. residuals, (c) confidence interval width vs. median predicted value, (d) predicted value distribution and confidence interval.

Heat Flux: the Katto correlation and the Bowring correlation. These correlations, derived from extensive experimental datasets, serve as established benchmarks in the thermal-hydraulics and nuclear engineering communities. This comparison aims to quantitatively assess the ML model's accuracy relative to these established correlations, identify any systematic biases or trends, and determine the model's suitability for predicting CHF within the relevant parameter space. A brief description of the correlations used for comparison in the study are provided in Table 7.

The visual comparison in Fig. 19 and the statistical metrics in Table 8 provide several key insights into the performance of the ML model relative to the established empirical correlations. Overall, within the domain of our comprehensive dataset, the ML model (RMSE = 10.5%, MAE = 7.4%) demonstrates a superior predictive capability compared to the Bowring (RMSE = 106.4%, MAE = 141.5%), Katto (RMSE = 52.8%, MAE = 89.6%), and Shah (RMSE = 77.6%, MAE = 134.3%) correlations. The scatter plot (Fig. 19) visually confirms this, with the ML model's predictions clustering more closely around the line of perfect agreement than any of the empirical correlations across a broad range of CHF values. Thus, the ML model's relatively consistent performance across a wider range of CHF values remains a significant advantage. Its ability to learn complex, non-linear relationships from the data allows it to adapt to the specific characteristics of the dataset, leading to better overall predictions.

However, a more nuanced analysis is crucial. A notable observation, evident in Fig. 19 and acknowledged in our initial analysis, is that the Bowring, Katto, and especially the Shah correlations, can exhibit relatively good agreement with experimental data at very low CHF values (below approximately 1–2 MW/m²). This suggests that the specific

formulations or underlying physical assumptions and simplifications made in their derivation might be particularly well-suited or more valid for CHF conditions under low heat flux or near-saturated conditions. For instance, the mechanisms governing CHF in these regimes might be simpler and more adequately captured by the explicit structure of these correlations.

Therefore, the choice between an ML model and an empirical correlation is not always straightforward and depends on the specific application, the available data, the required accuracy, the importance of interpretability, and the operational domain. While our ML model offers better overall predictive accuracy across a broader range of conditions within our dataset, empirical correlations retain distinct practical value in specific contexts. They remain particularly useful for scenarios involving very low critical heat flux (CHF) values where their performance aligns well with observations. These traditional correlations also serve as pragmatic alternatives when insufficient data exists to train reliable ML models, when rapid preliminary estimations are prioritized over computational complexity, or when seeking moderately stable (albeit potentially less accurate) extrapolations beyond the exact boundaries of the ML model's training domain. Their enduring relevance lies in balancing simplicity, transparency, and cautious applicability across these specialized use cases.

The findings underscore the potential of machine learning for enhancing CHF prediction accuracy but also emphasize the need for careful consideration of its limitations relative to established empirical correlations. Future work should continue to focus on expanding and diversifying the ML model's training dataset, exploring alternative model architectures, and conducting rigorous validation on independent datasets, particularly those covering operating conditions where current ML models might falter or where correlations excel.

Table 7
CHF empirical correlations.

Name	Formula	Parameter range
Katto [16]	$q_c = q_{c0} \times \left(1 + K \frac{\Delta h_i}{h_{fg}}\right)$	
	$\frac{q_{c0}}{Gh_{fg}} = \min \left[\frac{CDW^{0.043}}{0.1 \left(\frac{\rho_v}{\rho_l}\right)^{0.13} \times W^{1/3} \times \frac{D}{D+0.0031L}}, \frac{0.095 \times \left(\frac{\rho_v}{\rho_l}\right)^{0.133} \times W^{0.433}}{\left(\frac{L}{D}\right)^{0.27} \times \frac{D}{D+0.0031L}} \right]$	$\frac{\rho_v}{\rho_l} : 0.0003 \sim 0.41$ $D : 1 \sim 80 \text{ mm}$ $L : 0.01 \sim 8.8 \text{ m}$ $W : 3 \times 10^{-9} \sim 2 \times 10^{-2}$
	K is a dimensionless parameter,	
	$K = \min \left[\frac{0.261}{C} \times \left(\frac{G^2 L}{\rho_l \sigma}\right)^{0.043}, \frac{5}{6} \times \frac{0.0124L + D}{W^{1/3} L} \times \left(\frac{\rho_l}{\rho_v}\right)^{0.133} \right]$	
Bowring [17]	$W = \frac{\rho_l \sigma}{G^2 L}, \quad C = 0.34$	
	$q_{CHF} = \frac{A' + 0.25 DG \Delta h_{sub}}{C' + L}, \quad A' = \frac{2.317(0.25 h_{fg} DG) F_1}{1 + 0.0143 F_2 D^{1/2} G}$	
	$C' = \frac{0.077 F_3 DG}{1 + 0.347 F_4 \frac{G}{1356}}, \quad n = 2.0 - 0.5 p_r, \quad p_r = p/6.895$	
	when $p_r < 1$ $\left\{ \begin{array}{l} F_1 = \frac{p_r^{18.942} \exp[20.8(1 - p_r)] + 0.917}{1.917} \\ F_2 = \frac{p_r^{1.316} \exp[2.444(1 - p_r)] + 0.309}{1.309} \\ F_3 = \frac{p_r^{17.023} \exp[16.658(1 - p_r)] + 0.667}{1.667} \\ F_4 = \frac{p_r^{1.649}}{p_r^{1.649}} \end{array} \right.$	$p : 0.2 \sim 19.0 \text{ MPa}$ $D : 2 \sim 45 \text{ mm}$ $L : 0.15 \sim 3.7 \text{ m}$ $G : 136 \sim 18,600 \text{ kg/m}^2\text{s}$
Shah [41]	when $p_r > 1$ $\left\{ \begin{array}{l} F_1 = p_r^{-0.368} \exp[0.648(1 - p_r)] \\ F_2 = p_r^{-0.448} \exp[0.245(1 - p_r)] \\ F_3 = p_r^{0.219} \\ F_4 = \frac{p_r^{1.649}}{p_r^{1.649}} \end{array} \right.$	
	$Bo = \frac{q_{CHF}}{Gh_{fg}} = 0.124 \left(\frac{D}{L_E}\right)^{0.89} \left(\frac{10^4}{Y}\right)^n (1 - X_{IE})$	
	For n , If $Y \leq 10^4, n = 0$ If $Y \geq 10^4$, then	
	$n = \left\{ \begin{array}{l} \left(\frac{D}{L_E}\right)^{0.54} Y \leq 10^6 \\ \frac{0.12}{(1 - X_{IE})^{0.5}} Y > 10^6 \end{array} \right.$	$p : 0.014 \sim 0.96 \text{ MPa}$ $D : 0.315 \sim 37.5 \text{ mm}$ $L/D : 1.3 \sim 940$ $G : 4 \sim 29,051 \text{ kg/m}^2\text{s}$ $x_{in} : -4 \sim +0.85$
	For effective length and effective inlet quality $X_{IN} = \frac{h_{in} - h_{sat}}{h_{fg}}$ If $X_{IN} \leq 0, L_E = L_c, X_{IE} = X_{IN}$ If $X_{IN} \geq 0, L_E = L_B, X_{IE} = 0$ For boiling length L_B $\frac{L_B}{D} = \frac{X_c}{4Bo} = \frac{L_c}{D} + \frac{X_{IN}}{4Bo}$ $Y = G^{1.8} D^{0.6} \left(\frac{C_p}{k \rho_L^{0.8} g^{0.4}}\right) \left(\frac{\mu_L}{\mu_G}\right)^{0.6}$	

6. Conclusion

This study focuses on predicting the critical heat flux in the flow boiling process of water within an annular channel. An experimental loop was established to obtain CHF data points that, together with experimental data from previous research, formed the database used to train the machine learning models. The comparison of the selected models provides a clear overview of the prediction accuracy and performance. The primary conclusions are:

1. Five critical heat flux prediction models based on machine learning algorithms were developed, all using hydraulic diameter, heating length, pressure, mass flux, and inlet subcooling as input

Table 8

Statistical analysis of CHF data between ML and empirical correlations.

Correlations	RMSE	MAE
ML	10.5%	7.4%
Katto	52.8%	89.6%
Bowring	106.4%	141.5%
Shah	77.6%	134.3%

features. The CHF was set as the output parameter. The five machine learning models evaluated in this study all exhibited strong predictive capabilities for CHF, as demonstrated by their RMSE

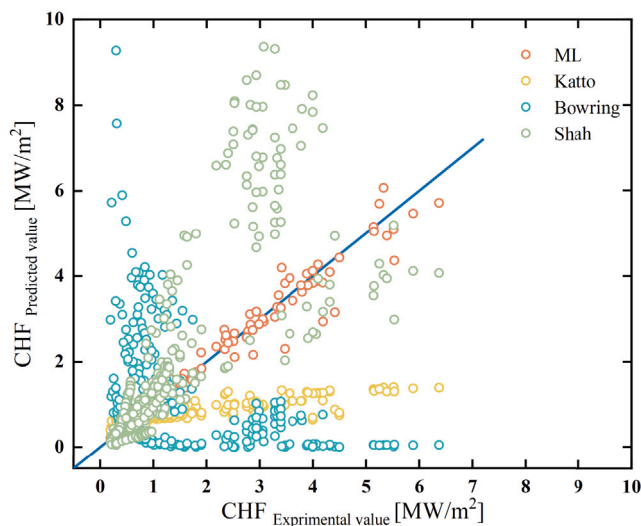


Fig. 19. Sensitivity analysis results.

values of 0.2819 to 0.1856, MAE values of 0.1541 to 0.1029, and R2 values of 0.9414 to 0.9753. Among them, the SVM model achieved the best prediction performance and highest accuracy with a RMSE of 0.1856 and an R2 of 0.9753.

2. Cross-validation was utilized for dataset partitioning and validation. By comparing the prediction accuracy of different dataset partitioning methods, 10-fold cross-validation was found to yield the best performance. This indicates that cross-validation can effectively improve the prediction performance and generalization ability of the model when dealing with medium-sized datasets, overcoming the limitations and specificity of traditional fixed dataset partitioning methods.
3. To determine the comparative importance of the selected features in CHF prediction, a parametric effects analysis was performed. Both the feature importance and sensitivity analysis show that mass flux consistently ranks as the most significant feature across all analysis methods, indicating its importance in the prediction of CHF. Due to the inconsistent importance of the remaining features, further research is needed to explore the deeper relationships between them and CHF. Meanwhile, the analysis results can better assist in the selection of feature parameters in future CHF prediction studies and further CHF enhancement.
4. The feature importance analysis and sensitivity analysis demonstrates that machine learning predictions align with established CHF dynamics - quantifying mass flux as the dominant parameter and revealing nonlinear dependencies. Secondary influence of inlet subcooling on CHF onset is also demonstrated. Examination of trained models reveals that synergistic interactions between mass flux, pressure, and subcooling significantly influence CHF. These observations suggest that enhanced liquid delivery to the heated surface, driven by the increased convective heat transfer and potentially augmented by subcooling-induced density differences, is crucial for delaying the boiling crisis. This bridges data-driven models with classical thermo-hydraulic theories, advancing mechanistic interpretations of CHF in annular flows.
5. The implemented uncertainty quantification, utilizing a Bagging SVM ensemble and quantile analysis, successfully generated well-calibrated 90% prediction intervals, achieving a Prediction Interval Coverage Probability of 90.6%. This demonstrates the method's reliability in capturing the true CHF values within the predicted bounds. The analysis further revealed that the

model generally assigns narrower intervals (higher confidence) to predictions of lower CHF values and appropriately widens these intervals for higher CHF magnitudes or when prediction errors are larger. Overall, these UQ findings significantly enhance the trustworthiness of the ML model, providing crucial uncertainty awareness that is vital for its practical application in safety-critical CHF assessments.

CRedit authorship contribution statement

Haidong Liu: Validation, Formal analysis, Data curation, Conceptualization. **Hongfu Jiang:** Writing – review & editing, Writing – original draft, Conceptualization. **Deqi Chen:** Supervision, Resources, Funding acquisition. **Jiang Qin:** Investigation. **Yangyang Wang:** Validation, Methodology. **Mingxia Liu:** Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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